

Dissertation Abstract

Physical inspection and root-cause failure analysis of three-dimensional integrated circuits (3D ICs) are essential for reliability assurance in advanced semiconductor packaging. The buried interfaces and heterogeneous material stacks characteristic of these devices complicate defect localization and limit the effectiveness of conventional inspection techniques, particularly for architectures incorporating TSVs, micro-bumps, hybrid bonding, and redistribution layers. No single modality offers sufficient sensitivity and coverage across the range of defect classes and length scales that arise in these stacks, motivating intelligent multimodal inspection frameworks that integrate X-ray, acoustic, terahertz, and optical modalities to reveal internal discontinuities and bonding defects. This dissertation examines the benefits and challenges of current multimodal inspection pipelines, emphasizing their potential to automate root-cause identification while reducing reliance on destructive validation. Key issues such as inter-modal registration errors, inconsistent contrast mechanisms, and data heterogeneity remain unresolved and presently limit scalability and interpretability. To address these limitations, the work develops artificial-intelligence-based fusion strategies that integrate multimodal information at the feature and decision levels and enhance sensitivity to hidden defects that are ambiguous or invisible in any single modality. The framework is evaluated on representative 3D IC samples and benchmarked against single-modality baselines to quantify the gains attributable to fusion.

The proposed approach advances non-destructive analysis and supports more reliable failure attribution in next-generation 3D IC systems, contributing methodologically through multimodal fusion strategies designed for the specific physics of packaging defects and practically through inspection pipelines suitable for adoption in failure-analysis laboratories and national-laboratory metrology programs.